

Conditional Polynomial Dimension from a Finite SQ Output Catalog

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1 Theoretical Setup

1.1 Problem Setting

Let X be an arbitrary domain, possibly empty, equipped with its power-set sigma algebra, and let $H \subseteq \{+1, -1\}^X$ be nonempty. All random variables and protocol selectors below are measurable. Since the functions being integrated are bounded, every displayed expectation is well defined. For a distribution D on X , a target $h \in H$, and a binary predictor $g : X \rightarrow \{+1, -1\}$, define

$$\mathcal{L}_{D,h}(g) := D\{x : g(x) \neq h(x)\} = \frac{1 - \mathbb{E}_{x \sim D}[h(x)g(x)]}{2}. \quad (1.1)$$

The deterministic dimension complexity is

$$\text{dc}(H) := \inf \left\{ d \in \mathbb{N}_0 : \begin{array}{l} \exists \phi : X \rightarrow \mathbb{R}^d, \quad \forall h \in H \exists w_h \in \mathbb{R}^d, \\ \forall x \in X, \quad h(x) \langle w_h, \phi(x) \rangle > 0 \end{array} \right\}, \quad (1.2)$$

where $\inf \emptyset := +\infty$. Thus one deterministic map must work for the entire class, and its signs must be strict at every point of the original domain. This is neither an average nor a probabilistic, distribution-dependent, or approximate representation.

Fix

$$m \in \mathbb{N}_0, \quad \tau > 0, \quad 0 \leq \varepsilon < \frac{1}{4}, \quad \rho := 1 - 2\varepsilon \in \left(\frac{1}{2}, 1\right],$$

together with catalog-family constants $B \geq 1$ and $k \in \mathbb{N}$, $k \geq 1$. A randomized learner A has a random tape U with fixed law ν and is deterministic conditional on $U = u$. At a nonterminal round $t \leq m$, after the full real reply transcript $v_{<t} \in \mathbb{R}^{t-1}$, it issues a query

$$q_{u,v_{<t}} : X \times \{+1, -1\} \longrightarrow [-1, 1]$$

that may depend arbitrarily on u and $v_{<t}$. It may stop at any depth $T \leq m$, including $T = 0$, and then returns a binary predictor. Every query has the canonical decomposition

$$a_q(x) := \frac{q(x, +1) + q(x, -1)}{2}, \quad b_q(x) := \frac{q(x, +1) - q(x, -1)}{2},$$

so that $q(x, y) = a_q(x) + yb_q(x)$ and $|a_q(x)| + |b_q(x)| \leq 1$. Neither component is restricted; in particular, label-independent query components are allowed.

For D, h, q , define the population center and full tolerance interval by

$$\mu_q(D, h) := \mathbb{E}_{x \sim D} q(x, h(x)), \quad I_q(D, h) := [\mu_q(D, h) - \tau, \mu_q(D, h) + \tau].$$

A valid adaptive policy $\pi \in \Pi(D, h)$ is nonanticipating. After observing all preceding query-response pairs and the currently issued query q_t , it may choose any real response

$$\begin{aligned} v_t &= \pi_t(q_1, v_1, \dots, q_{t-1}, v_{t-1}, q_t), \\ v_t &\in I_{q_t}(D, h). \end{aligned}$$

The policy may react to every revealed randomized query but not to unrevealed future learner coins. A randomized oracle is covered tape by tape after its own seed is fixed, so the accuracy premise below averages only over learner coins. No response alphabet, grid, exact-expectation response, or favorable policy is imposed. Write $e(u; D, h, \pi)$ for the complete execution and $\hat{h}_{u; D, h, \pi}$ for its terminal predictor.

1.2 Assumptions

Assumption 1 (Boundary-inclusive source regime). *The parameters satisfy $m \in \mathbb{N}_0$, every finite $\tau > 0$ is allowed, and $0 \leq \varepsilon < 1/4$. In particular, zero-query protocols, exact learning ($\varepsilon = 0$), and large tolerances are included; no condition $m \geq 1$, $\tau \leq 1$, or $\varepsilon > 0$ is imposed.*

Assumption 2 (One randomized adaptive learner). *The same learner A , fixed before D and h , uses $U \sim \nu$, may choose each query from the random tape and the entire preceding real-valued reply transcript, and terminates after at most m queries. It is not changed after observing the distribution, target, or oracle policy.*

Assumption 3 (Bounded unrestricted SQ access). *Every issued query is an arbitrary map $q : X \times \{+1, -1\} \rightarrow [-1, 1]$. No correlational-SQ, label-linearity, finite-dictionary, nonadaptivity, deterministic-query, finite-encoding, or response-tree-size restriction is imposed.*

Assumption 4 (All continuous tolerance-valid replies). *For every D and h , the learner is evaluated against every nonanticipating policy in $\Pi(D, h)$, with every response chosen arbitrarily from the full real interval $I_q(D, h)$. Exact population responses and favorable response selection are not assumed.*

Assumption 5 (Static terminal-output factorization). *The learner specification exposes one finite catalog $G = \{g_1, \dots, g_L\} \subseteq \{+1, -1\}^X$ with $L \geq 1$, fixed before D, h , the reply policy, all replies, and the sampled learner tape. On every complete tolerance-valid execution there is an explicit measurable selector J such that*

$$\hat{h}_{u; D, h, \pi} = g_{J(e(u; D, h, \pi))} \quad \text{for every } D, h, \pi, u.$$

The selector may factor an uncountable continuous transcript tree through $[L]$; neither a finite response alphabet nor finitely many transcript leaves is required. The indexed members $g_1, \dots, g_L$ are distinct functions.

Assumption 6 (Distribution-target-policy uniform accuracy). *For every distribution D on X , every $h \in H$, and every $\pi \in \Pi(D, h)$,*

$$\mathbb{E}_{u \sim \nu} \mathcal{L}_{D, h}(\hat{h}_{u; D, h, \pi}) \leq \varepsilon.$$

The displayed expectation is only over the learner tape; distributions, targets, and tolerance-valid adversarial reply policies are universally quantified.

Assumption 7 (Boundary-corrected polynomial output budget). *The catalog size obeys*

$$1 \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k.$$

The constants $B \geq 1$ and integer $k \geq 1$ are fixed for the designated learner family and are independent of $X, H, m, \tau, \varepsilon, D, h$, the reply policy and replies, and the learner coins. The bound has no hidden constants or suppressed parameter dependence.

The catalog may contain functions outside H ; it constrains terminal outputs and is not itself a representation hypothesis about H . The proof will derive, rather than assume, two facts. First, for every D, h, π , the output law on G has probabilities

$$p_i(D, h, \pi) := \nu\{u : \hat{h}_{u;D,h,\pi} = g_i\}$$

whose mixture correlation is at least ρ , and hence some catalog coordinate has correlation at least ρ . Second, for every finite $F \subseteq X$, the set

$$K_{h,F} := \left\{ w \in \Delta_L : h(x) \sum_{i=1}^L w_i g_i(x) \geq \rho \text{ for every } x \in F \right\}$$

is nonempty, and these sets have the finite-intersection property in the one fixed simplex Δ_L . These are proof obligations, not additional theorem-facing assumptions.

1.3 Formalized Goal

The goal is the following exact conditional conclusion. The one deterministic catalog map

$$\phi_G : X \rightarrow \mathbb{R}^L, \quad \phi_G(x) = (g_1(x), \dots, g_L(x)),$$

must satisfy

$$\forall h \in H \exists w_h \in \Delta_L \forall x \in X, \quad h(x) \langle w_h, \phi_G(x) \rangle \geq \rho = 1 - 2\varepsilon > \frac{1}{2}.$$

The map is independent of D, h , every valid reply policy and transcript, and the learner tape. The weight w_h may depend only on h , the fixed catalog, and the fixed accuracy parameter. Exact pointwise signs and

$$\text{dc}(H) \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k$$

must follow deterministically on every domain. The conclusion includes $m = 0$, every finite $\tau > 0$, $L = 1$, $B = 1$, $\varepsilon = 0$, and the empty domain. Its progress type is conditional: it depends on Assumptions 5 and 7. It neither asserts the boundary-invalid rate Cm/τ^2 nor derives a finite catalog for an arbitrary unrestricted response tree. Removing the catalog condition while proving a boundary-corrected catalog-free linear bound remains open.

2 Preliminaries

For the fixed catalog in Assumption 5, let

$$[L] := \{1, \dots, L\}, \quad \Delta_L := \left\{ w \in \mathbb{R}_+^L : \sum_{i=1}^L w_i = 1 \right\}.$$

The simplex records mixtures of terminal catalog functions and supplies the target-dependent vectors in the main theorem. Define the catalog-coordinate map and its mixture score by

$$\phi_G(x) := (g_1(x), \dots, g_L(x)) \in \mathbb{R}^L, \quad s_w(x) := \langle w, \phi_G(x) \rangle = \sum_{i=1}^L w_i g_i(x).$$

This notation is derived solely from the fixed catalog. The map ϕ_G is the common map in Theorem 3.1, while s_w is used to state its signed-margin conclusion. All other helper objects are introduced only inside the proofs where they are needed.

3 Main Theorem

Theorem 3.1 (Conditional polynomial catalog theorem). *Under Assumptions 1–7, the single deterministic catalog map $\phi_G : X \rightarrow \mathbb{R}^L$ satisfies*

$$\forall h \in H \ \exists w_h \in \Delta_L \ \forall x \in X, \quad h(x) \langle w_h, \phi_G(x) \rangle \geq 1 - 2\varepsilon = \rho > \frac{1}{2}.$$

The map is fixed independently of D, h , every valid reply policy, all replies and transcripts, and the learner tape. Each w_h may depend only on h, G , and ε . Consequently the signs are exact pointwise and

$$\text{dc}(H) \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k.$$

The conclusion is deterministic, fixed-horizon for executions of depth at most m , and uses the exact pointwise signed inner product and the dimension notion in (1.2). Its exposed quantities are $\varepsilon, \rho, L, B, k, m, \tau$, and $\text{dc}(H)$; there are no hidden constants. The constants B, k are fixed exactly as in Assumption 7, and no asymptotic limit or probability conversion is taken. For each theorem instance, the learner specification and its catalog are fixed quantities, while m, τ, ε , and L remain exposed as displayed.

At $\varepsilon = 0$ the pointwise signed margin is exactly 1. At $m = 0$ the bound is $\text{dc}(H) \leq L \leq B$ for every finite $\tau > 0$. The same statement retains the unique simplex weight when $L = 1$, the literal bound with no multiplier when $B = 1$, and $\text{dc}(H) = 0$ when $X = \emptyset$.

Theorem 3.1 is conditional on the static finite-output factorization and its polynomial budget. It does not imply that an arbitrary unrestricted adaptive SQ response tree admits such a catalog. Establishing a boundary-corrected catalog-free linear dimension bound from m and τ alone is the remaining full source gap.

4 Proof Sketch

For a fixed distribution, target, and arbitrary valid policy, push the learner tape through the measurable terminal selector. This produces an exact finite law on the catalog and exact loss and correlation mixtures, including at zero queries. Universal expected accuracy then yields a catalog-mixture correlation at least ρ , and a finite maximum dominates that mixture.

For a nonempty finite $F \subseteq X$, apply the policy-free correlation bound to every distribution supported on F . A self-contained finite minimax argument converts the resulting $\forall r \exists i$ statement into one $w \in \Delta_L$ satisfying every point constraint on F , without margin loss. The empty set is handled directly. The finite constraint sets are closed subsets of the fixed compact simplex, and their finite intersections are exactly the constraint sets associated with finite unions. Compactness therefore gives one w_h satisfying all constraints on an arbitrary domain.

Finally, the catalog score is exactly the inner product with ϕ_G . Because $\rho > 1/2$, its signed margin gives strict signs, and the resulting L -coordinate common map is an admissible witness in the definition of $\text{dc}(H)$. Appending the primitive catalog budget gives the stated dimension chain. The complete arguments appear in Propositions A.3, A.4, A.6, A.8, A.9, and A.10.

References

RUDIN, W. (1976). *Principles of Mathematical Analysis*. 3rd ed. McGraw-Hill, New York. [10](#), [11](#), [12](#), [14](#)

A Proof Details

A.1 Terminal Catalog Law under Adaptive Replies

Proposition A.1 (Validity of the exact-center reply policy). *Under Assumptions 1, 2, 3, and 4, if D is a distribution on X and $h \in H$, then the rule*

$$\pi_t^{\text{ctr}}(q_1, v_1, \dots, q_{t-1}, v_{t-1}, q_t) := \mu_{q_t}(D, h) = \mathbb{E}_{x \sim D} q_t(x, h(x)) \quad (\text{A.1})$$

at every issued round is a measurable nonanticipating member of $\Pi(D, h)$. Consequently $\Pi(D, h) \neq \emptyset$, including for a run that stops at $T = 0$.

Proof. Fix D and h . Since X has the power-set sigma algebra and $q_t : X \times \{+1, -1\} \rightarrow [-1, 1]$, the map $x \mapsto q_t(x, h(x))$ is measurable and bounded. Its expectation in (A.1) therefore exists and lies in $[-1, 1]$.

At round t , the rule uses only the fixed pair (D, h) and the currently revealed query q_t . It sees the tape only through the information already revealed in q_t and inspects no future learner coin, so it is nonanticipating under Assumption 4. The setup's measurability convention applies to this protocol selector; no topology, encoding, or finite sigma algebra on the query space is introduced.

For every issued query,

$$|\pi_t^{\text{ctr}} - \mu_{q_t}(D, h)| = 0 \leq \tau,$$

and hence

$$\pi_t^{\text{ctr}} = \mu_{q_t}(D, h) \in [\mu_{q_t}(D, h) - \tau, \mu_{q_t}(D, h) + \tau] = I_{q_t}(D, h).$$

The rule is defined at every history at which a query is issued, regardless of earlier real responses, and remains valid when subsequent queries depend arbitrarily on those responses. After termination no further condition is imposed. If $T = 0$, in particular if $m = 0$, there is no issued query and all reply-validity conditions are vacuous. Thus the same contingent rule belongs to $\Pi(D, h)$. $\square$

Lemma A.1 (Finite terminal-selector pushforward). *Under Assumptions 1, 2, 4, and 5, if D is a distribution on X , $h \in H$, and $\pi \in \Pi(D, h)$ is arbitrary, then*

$$Z_{D, h, \pi}(u) := J(e(u; D, h, \pi)) \in [L] \quad (\text{A.2})$$

is measurable. Its fibers form a measurable finite partition,

$$p_i(D, h, \pi) = \nu\{u : Z_{D, h, \pi}(u) = i\} = \nu\{u : \hat{h}_{u; D, h, \pi} = g_i\}, \quad p(D, h, \pi) \in \Delta_L,$$

and every bounded $f : G \rightarrow \mathbb{R}$ satisfies

$$\mathbb{E}_{u \sim \nu} f(\hat{h}_{u; D, h, \pi}) = \sum_{i=1}^L p_i(D, h, \pi) f(g_i). \quad (\text{A.3})$$

These conclusions allow full-interval continuous adaptive replies, randomized adaptive queries, variable stopping, and $T = 0$.

Proof. Fix D, h , and an arbitrary $\pi \in \Pi(D, h)$; this policy is not replaced by the center policy of Proposition A.1. The taped learner's query and termination variables are measurable functions of the tape and the measurable transcript already produced. Since π is a measurable nonanticipating selector, each next response is a measurable composition of the revealed history with π . Induction through at most m rounds makes $u \mapsto e(u; D, h, \pi)$ measurable. Early stopping causes no limiting issue because the stopping depth is one of $0, 1, \dots, m$.

The selector J in Assumption 5 is measurable on every complete valid execution. Thus (A.2) is a measurable map into the finite discrete set $[L]$. Put

$$E_i(D, h, \pi) := \{u : Z_{D,h,\pi}(u) = i\}.$$

The events E_i are measurable, pairwise disjoint, and their union is the whole learner-tape space. Therefore

$$p_i = \nu(E_i) \geq 0, \quad \sum_{i=1}^L p_i = \nu\left(\bigcup_{i=1}^L E_i\right) = 1, \quad (\text{A.4})$$

which proves $p(D, h, \pi) \in \Delta_L$.

On E_i , the exact selector identity gives

$$\widehat{h}_{u;D,h,\pi} = g_{J(e(u;D,h,\pi))} = g_i. \quad (\text{A.5})$$

Because $g_1, \dots, g_L$ are the distinct indexed members of the set G , the converse also holds: a terminal output equal to g_i has selected index i . This proves the asserted equality between selector fibers and output events.

For bounded $f : G \rightarrow \mathbb{R}$, equation (A.5) gives the pointwise simple-function identity

$$f(\widehat{h}_{u;D,h,\pi}) = \sum_{i=1}^L \mathbf{1}_{E_i}(u) f(g_i).$$

Integrating and using only linearity of the finite sum proves (A.3). No transcript conditioning or regular conditional distribution is required.

Replies may be continuous, and queries may contain label-independent components. Randomized and reply-adaptive future queries can change the fibers E_i and their masses, but they create neither an outcome outside G nor an approximation term. A tape that stops at $T = 0$ still supplies a complete empty-reply execution to J ; if $m = 0$, this holds for every tape. When $L = 1$, E_1 is the whole tape space and $p_1 = 1$. If the oracle uses an independent random seed, fixing it gives one deterministic policy. It is nonanticipating, so the preceding argument applies; no expectation other than $u \sim \nu$ has been introduced. $\square$

Proposition A.2 (Exact terminal loss and correlation mixtures). *Under Assumptions 2, 4, and 5, and the conclusion of Lemma A.1, if D is a distribution on X , $h \in H$, and $\pi \in \Pi(D, h)$, then*

$$\mathbb{E}_{u \sim \nu} \mathcal{L}_{D,h}(\widehat{h}_{u;D,h,\pi}) = \sum_{i=1}^L p_i(D, h, \pi) \mathcal{L}_{D,h}(g_i), \quad (\text{A.6})$$

$$\mathbb{E}_{u \sim \nu} \mathbb{E}_{x \sim D} [h(x) \widehat{h}_{u;D,h,\pi}(x)] = \sum_{i=1}^L p_i(D, h, \pi) \mathbb{E}_{x \sim D} [h(x) g_i(x)]. \quad (\text{A.7})$$

Equivalently,

$$\sum_{i=1}^L p_i \mathbb{E}_D [h g_i] = 1 - 2 \sum_{i=1}^L p_i \mathcal{L}_{D,h}(g_i) = 1 - 2 \mathbb{E}_u \mathcal{L}_{D,h}(\widehat{h}_{u;D,h,\pi}). \quad (\text{A.8})$$

Proof. The loss $g_i \mapsto \mathcal{L}_{D,h}(g_i)$ lies in $[0, 1]$. Lemma A.1, applied to this bounded functional, gives (A.6). Likewise, $g_i \mapsto \mathbb{E}_D [h g_i]$ lies in $[-1, 1]$, so the same lemma gives (A.7). The inner D -expectation is simply one bounded scalar for each of finitely many catalog functions; no conditioning on the continuous transcript or interchange of an infinite sum is involved.

Equivalently, equation (A.5) gives, for every $x \in X$,

$$\mathbb{E}_{u \sim \nu} \hat{h}_{u;D,h,\pi}(x) = \sum_{i=1}^L p_i(D, h, \pi) g_i(x).$$

Multiplying by $h(x)$ and integrating over D moves only a finite sum and again yields (A.7).

Finally, (1.1) gives $\mathbb{E}_D[hg_i] = 1 - 2\mathcal{L}_{D,h}(g_i)$. Multiply this equality by p_i , sum over i , use (A.4), and then use (A.6). This proves (A.8), with zero residual at every link. $\square$

Proposition A.3 (Exact finite terminal catalog law). *Under Assumptions 1–5, if D is a distribution on X and $h \in H$, then $\Pi(D, h) \neq \emptyset$. For every arbitrary $\pi \in \Pi(D, h)$, the vector $p(D, h, \pi)$ belongs to Δ_L and obeys the exact identities (A.6)–(A.8). The assertion holds for all continuous tolerance-valid adaptive replies and for every stopping depth, including $T = 0$.*

Proof. Proposition A.1 constructs a measurable nonanticipating policy whose response is the exact center of the currently issued query’s full real tolerance interval, proving $\Pi(D, h) \neq \emptyset$. This policy is used only for nonemptiness.

For an arbitrary $\pi \in \Pi(D, h)$, Lemma A.1 pushes ν through the measurable map $J \circ e$ and yields the setting-defined simplex vector and its exact finite functional identity. Proposition A.2 then supplies the loss and correlation specializations.

The calculation never enumerates replies or transcripts; arbitrary continuous adaptive replies alter only the selector events. The mixture policy remains arbitrary, so the center policy is not a favorable-policy substitution. Only learner-tape randomness is pushed forward. At $T = 0$ the empty-reply execution is complete and J still selects an index; at $m = 0$ every run has this form, and at $L = 1$ the unique mass is one. If $X = \emptyset$, there is no distribution on X , so this distribution-indexed proposition is vacuous rather than based on a nonexistent D . $\square$

A.2 Universal Catalog Correlation

For this subsection, write

$$c_i(D, h) := \mathbb{E}_{x \sim D}[h(x)g_i(x)], \quad i \in [L].$$

Each $c_i(D, h)$ belongs to $[-1, 1]$.

Lemma A.2 (Exact policywise catalog correlation). *Under Assumptions 1 and 6, and the conclusions of Lemma A.1 and Proposition A.2, if D is a distribution on X , $h \in H$, and $\pi \in \Pi(D, h)$ is arbitrary, then*

$$\sum_{i=1}^L p_i(D, h, \pi) c_i(D, h) = 1 - 2\mathbb{E}_{u \sim \nu} \mathcal{L}_{D,h}(\hat{h}_{u;D,h,\pi}) \geq \rho. \quad (\text{A.9})$$

The equality has zero residual, and its only algorithmic expectation is over the learner tape.

Proof. Fix arbitrary D, h , and $\pi \in \Pi(D, h)$. For every i and x ,

$$\mathbf{1}\{g_i(x) \neq h(x)\} = \frac{1 - h(x)g_i(x)}{2}.$$

Integration with respect to D gives

$$c_i(D, h) = 1 - 2\mathcal{L}_{D,h}(g_i). \quad (\text{A.10})$$

Using $p_i \geq 0$, $\sum_i p_i = 1$, and Proposition A.2, we obtain

$$\begin{aligned} \sum_{i=1}^L p_i(D, h, \pi) c_i(D, h) &= \sum_{i=1}^L p_i(D, h, \pi) (1 - 2\mathcal{L}_{D,h}(g_i)) \\ &= 1 - 2 \sum_{i=1}^L p_i(D, h, \pi) \mathcal{L}_{D,h}(g_i) \\ &= 1 - 2\mathbb{E}_{u \sim \nu} \mathcal{L}_{D,h}(\hat{h}_{u;D,h,\pi}). \end{aligned}$$

All sums are finite; no response approximation, transcript conditioning, oracle-randomness average, or expectation interchange occurs.

Assumption 6 applies to this same arbitrary policy. Multiplying its upper bound by -2 reverses the inequality, and therefore

$$1 - 2\mathbb{E}_{u \sim \nu} \mathcal{L}_{D,h}(\hat{h}_{u;D,h,\pi}) \geq 1 - 2\varepsilon = \rho.$$

This proves (A.9) for every valid policy. If $\varepsilon = 0$, then the left side is at least 1 and, as a convex average of numbers at most 1, is also at most 1; hence it equals 1 exactly. The proof uses no query round, so Proposition A.3 makes the calculation unchanged at $m = 0$ or $T = 0$. $\square$

Proposition A.4 (Policy-free best catalog correlation). *Under Assumptions 1 and 6, and the conclusions of Proposition A.1, Lemma A.1, and Lemma A.2, if D is a distribution on X and $h \in H$, then every $\pi \in \Pi(D, h)$ satisfies*

$$\max_{i \in [L]} c_i(D, h) \geq \sum_{i=1}^L p_i(D, h, \pi) c_i(D, h) \geq \rho. \quad (\text{A.11})$$

Consequently,

$$\max_{i \in [L]} \mathbb{E}_{x \sim D} [h(x) g_i(x)] \geq \rho \quad (\text{A.12})$$

for every D, h , independently of the valid policy and its output probabilities.

Proof. Fix D, h and first let $\pi \in \Pi(D, h)$ be arbitrary. Since $L \geq 1$, the finite maximum exists, and $c_i(D, h) \leq \max_{j \in [L]} c_j(D, h)$ for every i . Multiply these coordinate inequalities by the nonnegative simplex masses and sum:

$$\begin{aligned} \sum_{i=1}^L p_i(D, h, \pi) c_i(D, h) &\leq \sum_{i=1}^L p_i(D, h, \pi) \max_{j \in [L]} c_j(D, h) \\ &= \max_{j \in [L]} c_j(D, h). \end{aligned}$$

Lemma A.2 supplies the other inequality in (A.11) for this same arbitrary policy.

Proposition A.1 ensures that $\Pi(D, h) \neq \emptyset$, so the already established every-policy inequality has at least one instance. Its left side depends only on D, h , and the fixed catalog and contains neither a policy nor an output law. This proves (A.12). The center policy is used only to prevent vacuity; it never replaces the arbitrary policy in (A.9).

When $L = 1$, one has $p_1 = 1$ and (A.11) reduces to $c_1(D, h) \geq \rho$. At $\varepsilon = 0$, the maximum is both at least and at most 1, so it equals 1. At $m = 0$ or $T = 0$, the preceding catalog law still supplies both the simplex vector and policy nonemptiness, and no query update enters this proof. If $X = \emptyset$, no distribution exists and the conclusion is vacuous.

Thus the expected-error premise has been converted, for the same arbitrary continuous adaptive policy, into the exact mixture correlation and then into a policy-free catalog maximum. The level $\rho = 1 - 2\varepsilon$ is unchanged, no τ -dependent term is introduced, and no probability mode is converted. $\square$

A.3 Finite Simultaneous Feasibility

For each nonempty finite set F , define its probability simplex by

$$\Delta_F := \left\{ r = (r_x)_{x \in F} \in \mathbb{R}_+^F : \sum_{x \in F} r_x = 1 \right\}.$$

Thus the coordinates of every $r \in \Delta_F$ are nonnegative and sum to one.

For fixed $h \in H$ and nonempty finite $F \subseteq X$, define

$$A_{xi}^{h,F} := h(x)g_i(x), \quad x \in F, \quad i \in [L]. \quad (\text{A.13})$$

When h, F are fixed, write A for this matrix. For $w \in \Delta_L$ and $r \in \Delta_F$,

$$(Aw)_x = \sum_{i=1}^L w_i A_{xi}, \quad (r^\top A)_i = \sum_{x \in F} r_x A_{xi}.$$

Lemma A.3 (Finite-support catalog payoff). *Under Assumptions 5 and 6, and Proposition A.4, if $h \in H$, $\emptyset \neq F \subseteq X$ is finite, and $r \in \Delta_F$, define*

$$D_r(S) := \sum_{x \in F \cap S} r_x, \quad S \subseteq X. \quad (\text{A.14})$$

Then D_r is a distribution on X and

$$\max_{i \in [L]} \sum_{x \in F} r_x h(x) g_i(x) = \max_{i \in [L]} \mathbb{E}_{z \sim D_r} [h(z) g_i(z)] \geq \rho. \quad (\text{A.15})$$

Consequently,

$$\min_{r \in \Delta_F} \max_{i \in [L]} \sum_{x \in F} r_x h(x) g_i(x) \geq \rho, \quad (\text{A.16})$$

and the minimum is attained.

Proof. The masses in (A.14) are nonnegative, have total mass one, and have finite support, so the formula defines a countably additive probability measure on the power-set sigma algebra. For every i ,

$$\mathbb{E}_{z \sim D_r} [h(z) g_i(z)] = \sum_{x \in F} r_x h(x) g_i(x).$$

Proposition A.4, applied to this exact D_r and the fixed h , proves (A.15). The result is already policy-free; no reply policy, output-probability vector, or coherent maximizing index as r varies is required.

Since (A.15) holds for every r , its minimum is at least ρ . After enumerating F , the defining constraints make Δ_F a closed and bounded subset of $\mathbb{R}^{|F|}$, hence compact by the finite-dimensional Heine–Borel theorem (Rudin, 1976, Theorem 2.41). The objective is a finite maximum of linear functions of r and is continuous, so the extreme-value theorem (Rudin, 1976, Theorem 4.16) gives the asserted attained minimum. $\square$

Lemma A.4 (Finite convex-hull alternative). *Let F be a nonempty finite set, let $L \geq 1$, let $A \in \mathbb{R}^{F \times L}$, and let $\gamma \in \mathbb{R}$. If*

$$\forall r \in \Delta_F, \quad \max_{i \in [L]} \sum_{x \in F} r_x A_{xi} \geq \gamma, \quad (\text{A.17})$$

then there exists $w \in \Delta_L$ such that

$$\forall x \in F, \quad \sum_{i=1}^L w_i A_{xi} \geq \gamma. \quad (\text{A.18})$$

The threshold γ is unchanged.

Proof. The simplex Δ_L is closed and bounded in $\mathbb{R}^L$, hence compact by the finite-dimensional Heine–Borel theorem (Rudin, 1976, Theorem 2.41). The linear map $w \mapsto Aw$ is continuous, so its image is compact by preservation of compactness under continuous maps (Rudin, 1976, Theorem 4.14). Thus consider the compact convex set of column mixtures

$$C := \{Aw : w \in \Delta_L\} \subseteq \mathbb{R}^F$$

and the closed convex upper orthant

$$Q_\gamma := \{q \in \mathbb{R}^F : q_x \geq \gamma \text{ for every } x \in F\}.$$

Conclusion (A.18) is equivalent to $C \cap Q_\gamma \neq \emptyset$. Suppose instead that the two sets are disjoint.

For $a \in \mathbb{R}$, write $a_+ := \max\{a, 0\}$. Coordinatewise minimization over Q_γ gives, for every $c \in \mathbb{R}^F$,

$$\text{dist}(c, Q_\gamma)^2 = \sum_{x \in F} (\gamma - c_x)_+^2. \quad (\text{A.19})$$

The right side is continuous. Since C is nonempty and compact, the extreme-value theorem (Rudin, 1976, Theorem 4.16) gives a point $c^0 \in C$ minimizing this distance. Disjointness and (A.19) make the minimum strictly positive. Define

$$q_x^0 := \max\{c_x^0, \gamma\}, \quad y := q^0 - c^0, \quad \delta := \|y\|_2 > 0. \quad (\text{A.20})$$

Then $q^0 \in Q_\gamma$, $y_x = (\gamma - c_x^0)_+ \geq 0$, and $\delta = \text{dist}(c^0, Q_\gamma)$.

Fix $c \in C$. By convexity, $c_t := c^0 + t(c - c^0) \in C$ for $t \in [0, 1]$. Minimality of c^0 and $q^0 \in Q_\gamma$ imply

$$\delta \leq \text{dist}(c_t, Q_\gamma) \leq \|c_t - q^0\|_2.$$

Since $c^0 - q^0 = -y$, squaring and subtracting $\delta^2 = \|y\|_2^2$ yields

$$0 \leq -2t\langle y, c - c^0 \rangle + t^2\|c - c^0\|_2^2.$$

For $t > 0$, divide by t and let $t \downarrow 0$. Then

$$\langle y, c \rangle \leq \langle y, c^0 \rangle \quad \text{for every } c \in C. \quad (\text{A.21})$$

Set $S := \sum_{x \in F} y_x$. Since $y \geq 0$ and $y \neq 0$, one has $S > 0$ and $r^0 := y/S \in \Delta_F$. Whenever $y_x > 0$, definition (A.20) gives $q_x^0 = \gamma$; coordinates with $y_x = 0$ contribute nothing. Consequently,

$$\langle y, q^0 \rangle = \gamma S, \quad \langle y, c^0 \rangle = \gamma S - \|y\|_2^2. \quad (\text{A.22})$$

For the i th simplex vertex $e_i \in \Delta_L$, the column Ae_i lies in C . Equations (A.21) and (A.22) give, for every i ,

$$\sum_{x \in F} r_x^0 A_{xi} = \frac{\langle y, Ae_i \rangle}{S} \leq \frac{\langle y, c^0 \rangle}{S} = \gamma - \frac{\|y\|_2^2}{S} < \gamma.$$

The maximum over i is therefore below γ for $r^0 \in \Delta_F$, contradicting (A.17). Hence $C \cap Q_\gamma \neq \emptyset$, which is precisely (A.18). The proof reaches γ itself and uses no limiting sequence of smaller thresholds. $\square$

Proposition A.5 (Exact finite matrix minimax with attainment). *Let F be a nonempty finite set, let $L \geq 1$, and let $A \in \mathbb{R}^{F \times L}$. Then*

$$\begin{aligned} \min_{r \in \Delta_F} \max_{i \in [L]} \sum_{x \in F} r_x A_{xi} &= \max_{w \in \Delta_L} \min_{r \in \Delta_F} \sum_{x \in F} \sum_{i=1}^L r_x w_i A_{xi} \\ &= \max_{w \in \Delta_L} \min_{x \in F} \sum_{i=1}^L w_i A_{xi}. \end{aligned} \quad (\text{A.23})$$

The outer minimum and maximum are attained.

Proof. The functions

$$r \mapsto \max_{i \in [L]} \sum_{x \in F} r_x A_{xi}, \quad w \mapsto \min_{x \in F} \sum_{i=1}^L w_i A_{xi}$$

are finite maxima and minima of linear functions and hence continuous. The defining constraints make both nonempty finite simplices closed and bounded in their finite-dimensional coordinate spaces, hence compact by the finite-dimensional Heine–Borel theorem (Rudin, 1976, Theorem 2.41). The extreme-value theorem (Rudin, 1976, Theorem 4.16) therefore gives attainment. Let

$$\beta := \min_{r \in \Delta_F} \max_{i \in [L]} \sum_{x \in F} r_x A_{xi}, \quad \alpha := \max_{w \in \Delta_L} \min_{x \in F} \sum_{i=1}^L w_i A_{xi}.$$

For every $r \in \Delta_F$ and $w \in \Delta_L$,

$$\min_{x \in F} \sum_{i=1}^L w_i A_{xi} \leq \sum_{x \in F} \sum_{i=1}^L r_x w_i A_{xi} \leq \max_{i \in [L]} \sum_{x \in F} r_x A_{xi}. \quad (\text{A.24})$$

The middle term is a convex average of the row payoffs for fixed w and a convex average of the column payoffs for fixed r . Maximizing the left side and minimizing the right side gives $\alpha \leq \beta$.

By definition of β , every $r \in \Delta_F$ satisfies

$$\max_{i \in [L]} \sum_{x \in F} r_x A_{xi} \geq \beta.$$

Lemma A.4, applied with $\gamma = \beta$, gives $w^* \in \Delta_L$ for which

$$\min_{x \in F} \sum_{i=1}^L w_i^* A_{xi} \geq \beta.$$

Thus $\alpha \geq \beta$, so $\alpha = \beta$.

For fixed w , put $a_x := \sum_i w_i A_{xi}$. Every $r \in \Delta_F$ satisfies $\sum_x r_x a_x \geq \min_x a_x$, and equality holds at a simplex vertex supported on a minimizing row. Hence

$$\min_{r \in \Delta_F} \sum_{x \in F} \sum_{i=1}^L r_x w_i A_{xi} = \min_{x \in F} \sum_{i=1}^L w_i A_{xi}.$$

Likewise, for fixed r , maximizing the bilinear expression over Δ_L gives the largest pure-column payoff. These identities, together with $\alpha = \beta$, prove (A.23) in the stated row-minimizer/column-maximizer order. $\square$

Proposition A.6 (Exact finite catalog feasibility). *Under Assumptions 5 and 6, and Propositions A.4 and A.5, if $h \in H$ and $F \subseteq X$ is finite, then*

$$K_{h,F} := \left\{ w \in \Delta_L : h(x) \sum_{i=1}^L w_i g_i(x) \geq \rho \text{ for every } x \in F \right\}$$

is nonempty. If $F = \emptyset$, then $K_{h,F} = \Delta_L$. If $F \neq \emptyset$, then

$$\begin{aligned} \min_{r \in \Delta_F} \max_{i \in [L]} \sum_{x \in F} r_x h(x) g_i(x) &= \max_{w \in \Delta_L} \min_{r \in \Delta_F} \sum_{x \in F} \sum_{i=1}^L r_x w_i h(x) g_i(x) \\ &= \max_{w \in \Delta_L} \min_{x \in F} \sum_{i=1}^L w_i h(x) g_i(x) \geq \rho, \end{aligned} \quad (\text{A.25})$$

and an attained maximizer belongs to $K_{h,F}$.

Proof. If $F = \emptyset$, there are no constraints, so $K_{h,\emptyset} = \Delta_L$. Assumption 5 gives $L \geq 1$, and the first simplex vertex proves nonemptiness. No Δ_F , distribution on F , or minimax expression is formed in this case.

Suppose now that $F \neq \emptyset$. By Lemma A.3,

$$\min_{r \in \Delta_F} \max_{i \in [L]} \sum_{x \in F} r_x A_{xi} \geq \rho$$

for $A_{xi} = h(x)g_i(x)$. Proposition A.5 applies because both F and $[L]$ are nonempty and finite, and substituting this matrix proves (A.25), including its orientation and attainment.

Choose an attained maximizer $w_{h,F} \in \Delta_L$ in the last expression. Then

$$\min_{x \in F} \sum_{i=1}^L w_{h,F,i} h(x) g_i(x) \geq \rho,$$

so every $x \in F$ satisfies

$$h(x) s_{w_{h,F}}(x) = \sum_{i=1}^L w_{h,F,i} h(x) g_i(x) \geq \rho.$$

Thus $w_{h,F} \in K_{h,F}$. Every transfer is an equality or a direct consequence of an attained minimum, so the threshold remains exactly ρ . At $L = 1$, the maximizer is the unique simplex weight; at $\varepsilon = 0$, the same calculation retains margin 1.

For completeness, the logical chain is as follows. Every $r \in \Delta_F$ defines the genuine distribution D_r , and the policy-free correlation result yields the row-game value without selecting any policy or requiring the best index to be coherent in r . The reverse minimax inequality follows from Lemma A.4, and the attained finite-game mixture is already a vector in Δ_L . The proposition asserts no coherence among the separately produced $w_{h,F}$ for different finite sets; the next subsection supplies the arbitrary-domain passage. $\square$

A.4 Arbitrary-Domain Mixture Certificate

For $h \in H$ and finite $F \subseteq X$, retain the exact constraint set

$$K_{h,F} := \{ w \in \Delta_L : h(x) s_w(x) \geq \rho \text{ for every } x \in F \}. \quad (\text{A.26})$$

Lemma A.5 (Compact catalog simplex and closed exact constraints). *Under Assumption 5, Δ_L is a nonempty compact subset of $\mathbb{R}^L$. If $h \in H$ and $F \subseteq X$ is finite, then $K_{h,F}$ is closed in $\mathbb{R}^L$ and hence relatively closed in that same compact simplex. Moreover, $K_{h,\emptyset} = \Delta_L$.*

Proof. Assumption 5 gives a finite integer $L \geq 1$. The vector $e_1 = (1, 0, \dots, 0)$ belongs to Δ_L , so the simplex is nonempty. It has the representation

$$\Delta_L = \bigcap_{i=1}^L \{w \in \mathbb{R}^L : w_i \geq 0\} \cap \left\{ w \in \mathbb{R}^L : \sum_{i=1}^L w_i = 1 \right\}. \quad (\text{A.27})$$

Each set on the right is the inverse image of a closed subset of $\mathbb{R}$ under a continuous coordinate or sum functional. The closed-set form of the inverse-image characterization of continuity (Rudin, 1976, Theorem 4.8, by complements) therefore shows that Δ_L is closed. For $w \in \Delta_L$,

$$\|w\|_2 \leq \|w\|_1 = \sum_{i=1}^L w_i = 1,$$

so it is bounded. The finite-dimensional Heine–Borel theorem (Rudin, 1976, Theorem 2.41) therefore makes this one simplex compact.

Fix h, F , and $x \in F$. The finite linear functional

$$\lambda_{h,x}(w) := h(x) \sum_{i=1}^L w_i g_i(x)$$

is continuous. Since $[\rho, \infty)$ is closed, the closed-set form of the inverse-image characterization of continuity (Rudin, 1976, Theorem 4.8, by complements) makes its inverse image closed, and

$$K_{h,F} = \Delta_L \cap \bigcap_{x \in F} \lambda_{h,x}^{-1}([\rho, \infty)). \quad (\text{A.28})$$

The second intersection is finite, so $K_{h,F}$ is closed in $\mathbb{R}^L$. If $F = \emptyset$, it is an empty intersection equal to $\mathbb{R}^L$, and (A.28) gives $K_{h,\emptyset} = \Delta_L$ without introducing a point of X . $\square$

Proposition A.7 (Exact finite-intersection identity). *Under Assumption 5, Proposition A.6, and Lemma A.5, if $h \in H$, $n \in \mathbb{N}_0$, and $F_1, \dots, F_n \subseteq X$ are finite, then, with intersections taken inside Δ_L and with the nullary conventions*

$$\bigcap_{j=1}^0 K_{h,F_j} := \Delta_L, \quad \bigcup_{j=1}^0 F_j := \emptyset,$$

one has

$$\bigcap_{j=1}^n K_{h,F_j} = K_{h, \bigcup_{j=1}^n F_j} \neq \emptyset. \quad (\text{A.29})$$

Thus the family of all $K_{h,F}$, over finite $F \subseteq X$, has the finite-intersection property, including its empty finite subfamily.

Proof. For $n = 0$, the left side of (A.29) is Δ_L and the right side is $K_{h,\emptyset}$. They are equal by Lemma A.5 and nonempty because $e_1 \in \Delta_L$.

Now let $n \geq 1$ and put $F_\cup := \bigcup_{j=1}^n F_j$, which is finite. For $w \in \Delta_L$, the definition (A.26) gives

$$\begin{aligned} w \in \bigcap_{j=1}^n K_{h,F_j} &\iff h(x)s_w(x) \geq \rho \text{ for every } j \text{ and every } x \in F_j \\ &\iff h(x)s_w(x) \geq \rho \text{ for every } x \in F_\cup \\ &\iff w \in K_{h,F_\cup}. \end{aligned}$$

This proves the set identity. Proposition A.6 gives $K_{h,F_\cup} \neq \emptyset$. The same constraints and threshold occur on both sides, so neither coherence among finite witnesses nor a margin loss has been assumed. $\square$

Lemma A.6 (Arbitrary closed-family intersection on the catalog simplex). *Under Assumption 5 and Lemma A.5, let $\mathcal{A}$ be an arbitrary index set and let $\{C_\alpha\}_{\alpha \in \mathcal{A}}$ be a family of relatively closed subsets of Δ_L . If*

$$\bigcap_{\alpha \in \mathcal{A}_0} C_\alpha \neq \emptyset$$

for every finite $\mathcal{A}_0 \subseteq \mathcal{A}$, including the convention $\bigcap_{\alpha \in \emptyset} C_\alpha := \Delta_L$, then

$$\bigcap_{\alpha \in \mathcal{A}} C_\alpha \neq \emptyset.$$

Proof. If $\mathcal{A} = \emptyset$, the total intersection is the nonempty simplex Δ_L . Otherwise, suppose for contradiction that the total intersection is empty. The relatively open sets

$$O_\alpha := \Delta_L \setminus C_\alpha, \quad \alpha \in \mathcal{A},$$

then cover Δ_L . Compactness from Lemma A.5 gives a finite subcover indexed by some finite $\mathcal{A}_0 \subseteq \mathcal{A}$:

$$\Delta_L = \bigcup_{\alpha \in \mathcal{A}_0} O_\alpha.$$

Taking complements relative to Δ_L yields $\bigcap_{\alpha \in \mathcal{A}_0} C_\alpha = \emptyset$, contradicting the finite-intersection premise. The argument works for an index set of any cardinality and uses no sequence, measure, or topology on that index set. $\square$

Proposition A.8 (Exact arbitrary-domain catalog weight). *Under Assumption 5, Proposition A.6, Lemma A.5, Proposition A.7, and Lemma A.6, if $h \in H$, then there exists*

$$w_h \in \bigcap_{\substack{F \subseteq X \\ F \text{ finite}}} K_{h,F} \subseteq \Delta_L. \tag{A.30}$$

The same weight satisfies

$$\forall x \in X, \quad h(x)s_{w_h}(x) \geq \rho. \tag{A.31}$$

Both conclusions hold on the empty domain, and the margin is unchanged.

Proof. Fix $h \in H$. By Lemma A.5, every $K_{h,F}$ is relatively closed in the same compact simplex Δ_L . By Proposition A.7, every finite subfamily has nonempty intersection, including the empty subfamily. Apply Lemma A.6 with

$$\mathcal{A} := \{F \subseteq X : F \text{ is finite}\}, \quad C_F := K_{h,F}.$$

Its conclusion is the nonemptiness of the intersection in (A.30); choose one member and call it w_h .

For arbitrary $x \in X$, the singleton $\{x\}$ is finite. Equation (A.30) gives $w_h \in K_{h,\{x\}}$, and the exact definition of this set yields (A.31). Equivalently,

$$\rho - h(x)s_{w_h}(x) \leq 0 \quad \text{for every } x \in X,$$

so the finite-to-global passage has no limiting, approximation, or cardinality-dependent residual.

If $X = \emptyset$, the finite-subset index set contains only $\emptyset$ and the intersection is $K_{h,\emptyset} = \Delta_L \neq \emptyset$; one may choose e_1 , and (A.31) is vacuous. No distribution on the empty domain is formed. If $L = 1$, the construction returns the unique simplex weight, and if $\rho = 1$, every point constraint keeps margin 1.

The sets being intersected are determined only by h , the fixed catalog, and ρ . Thus w_h depends on no distribution, valid policy, reply, transcript, learner-tape realization, point x , or coherent choice of the finite-game witnesses. In particular, the exact identities (A.29) and the compact closed-family argument provide the full arbitrary-domain passage without an annihilating distribution or topology on X . $\square$

A.5 Strict Deterministic Catalog Representation

Lemma A.7 (Fixed catalog coordinates are the mixture score). *Under Assumption 5, the map*

$$\phi_G : X \longrightarrow \mathbb{R}^L, \quad \phi_G(x) = (g_1(x), \dots, g_L(x))$$

is determined solely by the fixed catalog. It is therefore independent of D, h , every valid policy and transcript, and the learner tape. For every $w \in \Delta_L$ and $x \in X$,

$$\langle w, \phi_G(x) \rangle = \sum_{i=1}^L w_i g_i(x) = s_w(x), \quad (\text{A.32})$$

and

$$|\langle w, \phi_G(x) \rangle| \leq 1. \quad (\text{A.33})$$

Proof. Assumption 5 fixes $g_1, \dots, g_L$ before any of D, h, π , the replies, or the learner tape is selected. Listing their deterministic values therefore introduces none of those later dependencies. Since each $g_i(x) \in \{+1, -1\}$, the listed vector lies in $\mathbb{R}^L$.

The Euclidean inner product with this vector is the finite coordinate sum in (A.32), which is exactly the definition of $s_w(x)$. The upstream score and the feature-map score are thus the same object, with no transform or residual. Finally, $w_i \geq 0$, $\sum_i w_i = 1$, and $|g_i(x)| = 1$, so

$$|\langle w, \phi_G(x) \rangle| \leq \sum_{i=1}^L w_i |g_i(x)| = \sum_{i=1}^L w_i = 1.$$

$\square$

Lemma A.8 (Strict binary signed-margin conversion). *Under Assumption 1, let $a \in \{+1, -1\}$ and $z \in \mathbb{R}$. If $az \geq \rho$, then*

$$\begin{cases} z \geq \rho > \frac{1}{2} > 0, & a = +1, \\ z \leq -\rho < -\frac{1}{2} < 0, & a = -1. \end{cases}$$

Consequently $az > 0$, $z \neq 0$, and the strict sign of z is the binary value a .

Proof. Assumption 1 gives $0 \leq \varepsilon < 1/4$, whence

$$2\varepsilon < \frac{1}{2}, \quad \rho = 1 - 2\varepsilon > \frac{1}{2} > 0.$$

For $a = +1$, the premise is $z \geq \rho$. For $a = -1$, it is $-z \geq \rho$, and multiplication by -1 reverses the inequality to $z \leq -\rho < -1/2 < 0$. The two cases exhaust the binary alphabet, and in both cases $az \geq \rho > 0$, excluding a zero score and any tie convention. $\square$

Proposition A.9 (Exact common-map catalog representation). *Under Assumptions 1 and 5, and Proposition A.8 and Lemmas A.7 and A.8, the single deterministic map ϕ_G satisfies*

$$\forall h \in H \exists w_h \in \Delta_L \forall x \in X, \quad h(x) \langle w_h, \phi_G(x) \rangle \geq \rho = 1 - 2\varepsilon > \frac{1}{2}. \quad (\text{A.34})$$

For the same w_h ,

$$\begin{aligned} h(x) = +1 &\implies \langle w_h, \phi_G(x) \rangle \geq \rho > 0, \\ h(x) = -1 &\implies \langle w_h, \phi_G(x) \rangle \leq -\rho < 0. \end{aligned}$$

Hence every signed product is strictly positive. The assertion retains margin 1 at $\varepsilon = 0$, the unique weight at $L = 1$, and vacuous pointwise truth when $X = \emptyset$.

Proof. Fix $h \in H$. Proposition A.8 gives one $w_h \in \Delta_L$ such that $h(x)s_{w_h}(x) \geq \rho$ simultaneously for all $x \in X$. Lemma A.7 gives, for each such x ,

$$h(x) \langle w_h, \phi_G(x) \rangle = h(x)s_{w_h}(x) \geq \rho. \quad (\text{A.35})$$

The parameter regime supplies $\rho > 1/2$, proving (A.34) without loss. Apply Lemma A.8 with $a = h(x)$ and $z = \langle w_h, \phi_G(x) \rangle$; its premise is exactly (A.35), and its two cases give the stated strict score signs.

Lemma A.7 also shows that ϕ_G is fixed once the primitive catalog is fixed and is common to all targets and protocol instances. Proposition A.8 constructs w_h from the constraint family determined by h, G , and ρ , so it depends on no distribution, policy, reply, transcript, tape realization, or queried point. The quantifier order is therefore $\forall h \exists w_h \forall x$.

If $\varepsilon = 0$, then $\rho = 1$ and (A.34) gives a signed product at least 1. Equation (A.33) and $|h(x)| = 1$ give the reverse bound, so the signed margin is exactly 1 pointwise. If $L = 1$, then $\Delta_1 = \{(1)\}$ and the same proof uses the unique weight and the one-coordinate map $(g_1(x))$. If $X = \emptyset$, the global weight still exists, ϕ_G is the unique function from the empty domain into $\mathbb{R}^L$, and all pointwise statements are vacuous; no distribution is formed. The live setting assumes $H \neq \emptyset$. If that restriction were dropped, the outer universal statement would be vacuous and the zero-dimensional map would give $\text{dc}(\emptyset) = 0$.

Thus the arbitrary-domain mixture score transfers to the primitive catalog inner product with zero residual, and strict positivity removes every sign tie. This is an exact deterministic L -coordinate representation. $\square$

A.6 Boundary-Corrected Dimension Bound

Proposition A.10 (Boundary-corrected catalog dimension bound). *Under Assumptions 1 and 7, the dimension definition (1.2), and Proposition A.9, one has*

$$\text{dc}(H) \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k. \quad (\text{A.36})$$

The conclusion is deterministic, fixed-horizon, and uses the exact strict pointwise sign representation. It holds without alteration at $m = 0$, for every finite $\tau > 0$, at $L = 1$, at $B = 1$, and on the empty domain. It is conditional on the catalog budget and makes no catalog-free or boundary-unadjusted Cm/τ^2 assertion.

Proof. Proposition A.9 supplies one deterministic map $\phi_G : X \rightarrow \mathbb{R}^L$ and, for every $h \in H$, one $w_h \in \Delta_L \subseteq \mathbb{R}^L$ such that

$$h(x)\langle w_h, \phi_G(x) \rangle \geq \rho > \frac{1}{2} > 0 \quad \text{for every } x \in X.$$

Since the finite positive integer L lies in $\mathbb{N}_0$, these objects make $d = L$ admissible in the set defining $\text{dc}(H)$ in (1.2). That set is nonempty, and its infimum satisfies

$$\text{dc}(H) \leq L. \tag{A.37}$$

The inference preserves the exact common-map quantifier order and makes no distributionwise, policywise, or pointwise selection of the feature map.

Assumption 7 states literally that

$$1 \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k. \tag{A.38}$$

The displayed $B \geq 1$ and integer $k \geq 1$ are fixed family constants, with exactly the independence recorded in that assumption. Appending the upper inequality in (A.38) to (A.37) proves (A.36) by transitivity, without a hidden constant or suppressed dependence.

The boundary regimes are direct specializations of the same inequalities. If $m = 0$, then every finite $\tau > 0$ gives

$$B \left(1 + \frac{0}{\tau^2}\right)^k = B, \quad \text{dc}(H) \leq L \leq B,$$

so the bound does not collapse to zero. For every finite $\tau > 0$, the denominator is positive and $1 + m/\tau^2 \geq 1$; no condition $\tau \leq 1$, large-tolerance limit, or deletion of the leading 1 occurs. If $L = 1$, Proposition A.9 gives $\text{dc}(H) \leq 1$, while (A.38) remains literal. If $B = 1$, then

$$\text{dc}(H) \leq L \leq \left(1 + \frac{m}{\tau^2}\right)^k.$$

In the joint case $B = 1$ and $m = 0$, the inequalities $1 \leq L \leq 1$ force $L = 1$.

If $X = \emptyset$, the unique map $X \rightarrow \mathbb{R}^0$, with the unique zero-dimensional vector for every $h \in H$, satisfies the pointwise condition vacuously. Hence $d = 0$ is admissible in (1.2); since all admissible dimensions are nonnegative, $\text{dc}(H) = 0$, and (A.36) reads $0 \leq L \leq B(1 + m/\tau^2)^k$. No distribution on the empty domain is used.

Only (A.37) follows from the strict representation alone. The second inequality uses the primitive budget (A.38); nothing here derives such a budget for an arbitrary unrestricted response tree. Nor is that expression replaced by Cm/τ^2 , which would contradict the $m = 0$ specialization. There is no auxiliary parameter, term absorption, probability conversion, horizon upgrade, or norm change. $\square$

A.7 Proof of the Main Theorem

Proof of Theorem 3.1. Proposition A.9, whose dependencies are ultimately the exact terminal catalog law in Proposition A.3, the universal correlation in Proposition A.4, the finite certificate in

Proposition A.6, and the arbitrary-domain passage in Proposition A.8, gives the single map ϕ_G and the exact pointwise margin

$$\forall h \in H \ \exists w_h \in \Delta_L \ \forall x \in X, \quad h(x) \langle w_h, \phi_G(x) \rangle \geq 1 - 2\varepsilon > \frac{1}{2}.$$

The same proposition supplies strict signs and the required independence of the map and weights. Proposition A.10 then applies the definition of $\text{dc}(H)$ and appends the primitive catalog budget to obtain

$$\text{dc}(H) \leq L \leq B \left(1 + \frac{m}{\tau^2}\right)^k.$$

Its proof also establishes the exact $m = 0$, finite- τ , $L = 1$, $B = 1$, and empty-domain specializations. These are precisely all conclusions of Theorem 3.1. $\square$